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SUPERSEDING
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MILITARY SPECIFICATION

WOOD PRESERVATIVE SOLUTIONS, OIL-SOLUBLE, SHIP AND BOAT USE

This specification has been approved by the Department of Defense, and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the ingredients, the solvents and the properties of oil-borne wood preservatives for the non-pressure treatment of ship and boat building lumber and plywood.

1.2 Classification. Oil-borne wood preservative shall be furnished ready-to-use and shall be of the following types, as specified (see 6.2):

Type A—Copper naphthenate.

Type B—Pentachlorophenol.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT-E-485	— Enamel, Semi-Gloss, Rust-Inhibiting.
TT-P-141	— Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
TT-T-291	— Thinner, Paint, Volatile Mineral Spirits (Petroleum - Spirits).
TT-W-570	— Wood - Preservative; Pentachlorophenol.
PPP-C-96	— Cans, Metal, 28-Gage and Lighter.
PPP-D-729	— Drums: Metal, 55-Gallon (for Shipment of Noncorrosive Materials).

FSC 8030

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PPP-D-760 — Drums and Pails, Metal (5 and 16.64 Gallon).

MILITARY

MIL-P-15130 — Paint, Outside, Haze-Gray, No. 27 (Formula No. 5H) (for Naval Shipboard Use).

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations—Title 49.

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

AMERICAN WOOD PRESERVER'S ASSOCIATION

A5-56—Standard Method for the Determination of Copper in Copper Naphthenates.

(Application for copies should be addressed to the Secretary-Treasurer, American Wood Preserver's Association, 839 - 17th St., N. W., Washington 6, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules

(Information as to the availability of these Regulations may be obtained from the Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Composition. The preservative solution shall consist of a toxic ingredient, a solvent and an anti-blooming agent (if necessary). For type A, the toxicant shall be copper naphthenate; for type B, the toxicant shall be pentachlorophenol conforming to Specification TT-W-570. The solvent shall conform to grade I of Specification TT-T-291. Suitable co-solvents may be added to increase solvency where necessary.

3.2 Finished formulation. When tested as specified in 4.4 the preservatives shall conform to table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.

4.1.1 *Lot.* For purposes of sampling, a lot shall be defined as the quantity of the wood preservative offered for Government inspection at one time from one manufacturing batch.

4.1.2 *Batch.* A batch shall consist of all material processed as a unit in quantities not exceeding a day's production or 2,000 gallons, whichever is smaller.

4.1.3 *Sampling for tests at a Government laboratory.* From each inspection lot, the inspector shall select two containers at random. From each of the two containers 1-quart specimens shall be taken and placed in separate, clean, dry, metal or glass containers and forwarded to the Laboratory Officer, Industrial Test Laboratory, Philadelphia Naval Shipyard, Naval Base, Philadelphia 12, Pennsylvania.

4.1.3.1 *Optional samples.* When requested by the contractor the inspector may select two 3-quart samples instead of two 1-quart samples. Each sample shall be divided into three parts. One part shall be forwarded to the testing laboratory in accordance with 4.1.3, one part to the contractor, and one part to be held by the inspector for use in case of dispute.

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TABLE I. *Finished formulation*

Property	Requirement		Applicable test paragraph
	Type A	Type B	
Copper naphthenate, as copper, gm./gal. (77°F.)	68 (min.)	.....	4.4.1.1
Pentachlorophenol, gm./gal. (77°F.)	.....	163 (min.)	4.4.1.2
Wax content	To contain no wax	To contain no wax	4.4.2
Flash point (°F.)	100 (min.)	100 (min.)	4.4.3
Insolubles, (ml./50 ml.)	0.1 (max.)	0.1 (max.)	4.4.4
Blooming	To be free from blooming and from deposits of crystals visible to the unaided eye.		4.4.5
Compatibility	Shall not interfere with application, drying or adhesion of paints, nor shall it bleed through such coatings.		4.4.5
Coloring	.....	When specified (see 6.2) shall contain a gray, blue, green or red dye which (after 48 hours drying) shall not interfere with paint performance.	.....

4.1.4 *Sampling for examination of containers.* A random sample of filled containers shall be selected by the inspector in accordance with Standard MIL–STD–105 at inspection level I and Acceptable Quality Level = 2.5 percent defective to verify compliance with all requirements of this specification regarding fill, closure, marking and other requirements not involving tests.

4.2 Examination.

4.2.1 *Examination of containers.* Each filled container selected as a sample shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample container shall also be weighed to determine the quantity of contents. Any container in the sample having one or more defects or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for Government inspection provided the contractor has removed or corrected all nonconforming containers.

4.3 Lot acceptance. The inspection test samples selected in accordance with 4.1.3 shall be subjected separately to the applicable tests specified in 4.4. The inspector shall not release a lot until a favorable report is received from the Government laboratory. Nonconformance of either sample shall be basis for rejection.

4.3.1 *Rejected lots.* Rejected lots may be offered again for Government inspection provided the contractor has removed or reworked all nonconforming products. The inspector shall again select and examine samples from such resubmitted lots to verify compliance with the requirements of this specification (see 4.1.3).

4.4 Tests.

4.4.1 *Toxic ingredients.* The wood preservative solution shall be tested to verify conformance with 3.2 as follows:

4.4.1.1 *Type A.* Pipette 5 ml. of the thoroughly mixed sample at 77°F. into a 250 ml. Kjeldahl flask. Add 25 ml. of concentrated sulfuric acid and heat gently (under hood) for ½ hour with occasional swirling. Care should be taken to avoid foaming over. In-

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crease the heat until sample is thoroughly charred and then add 2- to 3-ml. portions of concentrated nitric acid until the solution clears. Cool and transfer the solution to a 200-ml. tall form beaker. Wash the Kjeldahl flask with two 15 ml. portions of distilled water and add the washings to the acid solution. Heat the solution to boiling and boil for 3 minutes. Cool, carefully neutralize with ammonium hydroxide (caution—avoid spattering) until the solution turns dark blue and add a few drops in excess. If the solution is not clear at this point, filter through a number 40 Whatman paper. Wash paper well with distilled water, then discard paper and precipitate. At this stage, copper can be determined volumetrically or electrolytically. If the volumetric method is selected, make the above filtration into a 250-ml. Erlenmeyer flask and proceed in accordance with the method outlined in paragraph number 7, section A5-56 of the American Wood Preserver's Association Manual. If the electrolytic method is selected, proceed as follows: Make the above filtration into a 200-ml. tall form beaker, add 10 ml. (1:1) sulfuric acid and 1 ml. of concentrated nitric acid and dilute to approximately 120 ml. with distilled water. Proceed using the standard method for determining copper by electrolysis. The percent copper in the type A preservative shall be calculated as follows:

gms. copper per gal. = weight of copper x 757

4.4.1.2 *Type B.* Transfer a 5-ml. sample at 77°F. into a 250 ml. Erlenmeyer flask. Add 65 ml. of ethanol and gently swirl until solution of the sample is complete. Add 35 ml. of distilled water and 0.6 ml. of meta cresol purple indicator.[1] Titrate with aqueous N/10 NaOH to the first permanent purple end point. Run a blank on 65 ml. of ethanol and 35 ml. distilled water and subtract from above titration.

Calculations: Net ml. of 0.1000 NaOH x 2.11 = grams of pentachlorophenol per gallon

4.4.2 *Wax content.* Wax content shall be determined as follows:

4.4.2.1 *Types A and B.* Transfer 20 ml. of type A preservative to a 250 ml. Erlenmeyer flask, add 25 ml. of H_2SO_4 and heat on a steam bath for 1 hour with occasional agitation. Transfer to a 250 ml. separatory funnel, and after phases separate, discard acid (lower) layer. Add 25 ml. of 5 percent NaOH solution, shake thoroughly and discard caustic (lower) layer. Repeat with an additional 25 ml. of caustic, discarding caustic. Wash with 25 ml. of 5 percent HC1, discarding washings. Repeat with another 25 ml. of HC1. Transfer contents of separator to a crystallizing dish, evaporate to dryness on a steam bath and bake for 1 hour at 100 to 110°C. Cool and examine for wax, its presence being indicated by an opaque solid film. Disregard any oily film that appears on cooling.

4.4.3 *Flash point.* The flash point shall be determined in accordance with method 429.1 of Specification TT-P-141.

4.4.4 *Insolubles.* Fifty ml. of the preservative solution shall be placed in a 50-ml. centrifuge tube possessing a conical bottom and a graduated scale. The tube and contents shall be chilled to 0°F. overnight (16 hours minimum). The chilled tube shall be quickly placed in a centrifuge and shall be rotated for 3 minutes at 2000 r.p.m. The volume of sediment shall be read.

4.4.5 *Blooming and compatibility with paint.* One-half inch Douglas fir plywood panels (3 inches by 12 inches approximately) shall be immersed in the wood preservative to one-half their length for a period of 3 minutes. The panels shall then be hung vertically with the treated portion down for a period of 72 hours for type A preservative and 48 hours for type B at room temperature. At the end of this drying time, the treated surfaces shall be visually examined

[1] The indicator shall be prepared as follows. Place 0.100 gm. of meta cresol purple indicator in a small mortar; add 2.6 ml. of N/10 aqueous NaOH. Rub with the pestle until solution is complete. Transfer the solution to a 100-ml. volumetric flask and make up to volume with distilled water.

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for the absence of blooming, crystal deposits and glossy or waxy films. Upon completion of this examination, the panels shall be given one brush coat of exterior haze-gray paint conforming to Specification MIL-P-15130. The coating shall be allowed to dry for 24 hours at room temperature and examined for drying and adhesion over the treated and untreated halves of the specimen and also for bleeding of the wood preservative through the treated portion. If the paint is satisfactorily dry (a slight tackiness or softness being permissible for type A) and shows no evidence of bleeding of the wood preservative, a second coat of the haze-gray paint shall be applied to the test specimens. After drying 24 hours, the second coat shall be examined again for tackiness, adhesion and bleeding of preservative and no tackiness or softness shall be evident.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.* Wood preservative solution shall be furnished in 1 gallon can, 5 gallon or 55 gallon drum as specified (see 6.2).

5.1.1.1 *One gallon can.* The 1 gallon can shall conform to type V, class 2 of Specification PPP-C-96; plan B exterior coating will be required. When specified the containers shall be provided with either a wire handle or a formed bridge type handle securely affixed to the top. The wire used for the handle shall have a nominal diameter of not less than 0.1055 inch (12 gage).

5.1.1.2 *Five gallon drums.* The 5 gallon drums shall conform to type I of Specification PPP-D-760. Screw cap closures and inner seals shall be furnished (type I only). The drums shall be furnished with a flexible pouring spout. Exterior coating of drums will be required. Wire handles or bails shall be treated to resist corrosion. Unless otherwise specified the coating shall conform to Specification TT-E-485.

5.1.1.3 *Fifty-five gallon drums.* The 55 gallon drums shall conform to type I of Specification PPP-D-729. Cap seals shall be furnished on the drums. Exterior coating of the drums will be required. Unless otherwise specified the coating shall conform to Specification TT-E-485.

5.1.2 *Level C.* Wood preservative solution shall be packaged in accordance with the manufacturer's practice.

5.2 Packing.

5.2.1 *Level A and level B.* The material packaged in cans in accordance with level A or C as specified (see 6.2) shall be packed in accordance with appendix of Specification PPP-C-96. Five gallon or 55 gallon drums will require no additional packing.

5.2.2 *Level C.* The material packaged in accordance with level A or C as specified (see 6.2) shall be packed in containers of the type, size, and kind used for the purpose in a manner which will insure acceptance and safe delivery at destination. Shipping containers shall comply to the Consolidated Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Labeling.* The container shall be clearly labeled to show manufacturer's name, trademark, and type. The following precautionary marking shall be included on labels of individual containers:

> "Flammable; avoid fire hazard; AVOID skin contact or introducing into eyes. AVOID prolonged inhalation of fumes. Work area should be outdoors or well ventilated room. Use of goggles and protective clothing is recommended. Wash well with soap and water after using or handling this material."

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5.3.2 Oil-borne wood preservatives are "economic" poisons as defined by the Federal Insecticide, Fungicide and Rodenticide Act, and application for oil-borne wood preservative registration and labeling requirements under the Act should be made to the Insecticide Division, Livestock Branch, Production and Marketing Administration, Washington 25, D.C.

5.3.3 In addition to the marking specified in 5.3.1, 5.3.2 and any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with Standard MIL-STD-129. Precautionary markings shall be in accordance with the Code of Federal Regulations, Title 49.

6. NOTES

6.1 **Intended use.** The wood preservatives described by this specification are intended for use in preventing the decay of wood and to be compatible with subsequent gluing, caulking and painting operations. However, it is necessary that wood preservatives be allowed to dry before such subsequent operations are attempted. The usual drying time is 24 hours for a brush application and 72 hours for a 3-minute dip. If the wood preservative is not allowed to dry, the migration of the unevaporated solvent will discolor painted surfaces and prevent the satisfactory performance of glues and caulking compounds. Do not place treated lumber or plywood in solid piles as this prevents solvent evaporation. When applying wood preservatives, it is necessary to treat *all* surfaces. End grain surfaces are particularly susceptible to decay and should receive both thorough and repeated applications. The penetration of wood preservatives applied by non-pressure means is generally slight. Therefore, cut, bored, or sanded surfaces which have been previously treated require retreatment if adequate protection is to be assured. Surfaces to be varnished may be treated with type B preservative in which the coloring ingredients (see 6.2) have been omitted. Large vats of preservative solution should be covered when not in use. Otherwise solvent evaporation will result in concentrated solutions which can cause painting difficulty.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required (see 1.2).

(c) Whether coloring shall be added to type B (see 3.2).

(d) Size of container required (see 5.1.1).

(e) Level of packaging and packing required (see 5.1 and 5.2).

(f) That the material shall be purchased by volume, the unit being a U.S. gallon at 77°F. (25°C.). The volume shall be determined by dividing the net weight in pounds by weight per gallon. To obtain weight per gallon, multiply the specific gravity at 77/77° F. (25/25°C.) by 8.33.

(g) Type of handle, when required (see 5.1.1).

6.3 **Precautionary measures.**

6.3.1 *Fire hazard.* The solvent or thinner used in the wood preservative may have a flash point as low as 100°F. (37.8°C.), and care should be taken to prevent fire. No smoking, welding, or other source of fire should be permitted near the treating operation or treated material until after 24 hours of drying, under conditions of adequate ventilation to insure removal of inflammable vapors. Treating should be done out-of-doors or in an adequately ventilated room. The preservatives should be stored out-of-doors or in a "paint" room where adequate precautions are taken from fire hazard standpoint.

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6.3.2 *Irritating effects.* Although workmen vary in sensitivity, wood preservatives are more or less irritating. Adequate ventilation and other protective measures should be provided. The use of neoprene-coated gloves, apron coated with butyrol resin, and protective ointments are helpful. Workmen should be instructed to avoid the use of clothing soiled with the wood preservative and also to wash well with soap and water after contact with the solutions. Exposure of skin to direct sunlight should be avoided when using type B solution.

6.3.3 When brushing or dipping, adequate ventilation should be maintained to reduce the concentration of the solvent vapor.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Corps of Engineers
Navy—Bureau of Ships

Other interest:
Army—MQSigT
Navy—CG.

Preparing activity:
Navy—Bureau of Ships

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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